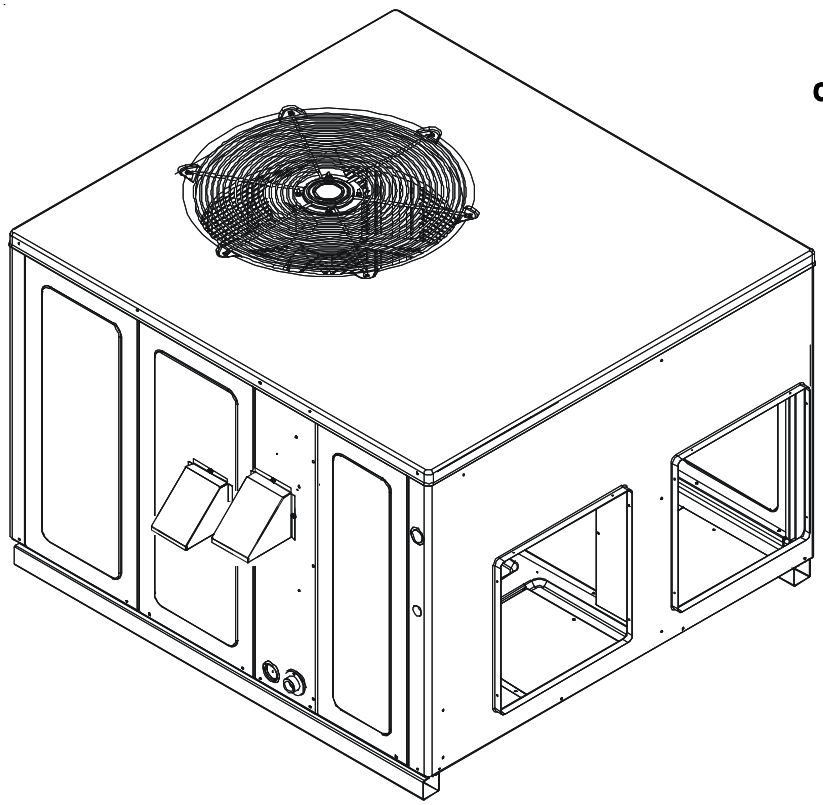
Service Instructions

APG/GPG 14 SEER
Gas Electric Package Units
with R-410A Refrigerant
& Accessories



This Forced Air Central Unit Design Complies With Requirements Embodied in The American National Standard / National Standard of Canada Shown Below.

ANSI Z21.47•CSA-2.3 Central Furnaces

This manual is to be used by qualified, professionally trained HVAC technicians only. Goodman does not assume any responsibility for property damage or personal injury due to improper service procedures or services performed by an unqualified person.

RS6300012r4
July 2017

Copyright © 2014-2017 Goodman Manufacturing Company, L.P.

SERVICING

COOLING ANALYSIS CHART

Complaint	No Cooling							Unsatisfactory Cooling					System Operating Pressures				Test Method Remedy	See Service Procedure Ref.
POSSIBLE CAUSE DOTS IN ANALYSIS GUIDE INDICATE "POSSIBLE CAUSE" SYMPTOM	System will not start	Compressor will not start - fan runs	Comp. and Cond. Fan will not start	Evaporator fan will not start	Condenser fan will not start	Compressor runs - goes off on overload	Compressor cycles on overload	System runs continuously - little cooling	Too cool and then too warm	Not cool enough on warm days	Certain areas too cool, others too warm	Compressor is noisy	Low suction pressure	Low head pressure	High suction pressure	High head pressure		
Pow er Failure	●																Test Voltage	S-1
Blow n Fuse	●		●	●													Inspect Fuse Size & Type	S-4
Loose Connection	●			●		●											Inspect Connection - Tighten	S-2
Shorted or Broken Wires	●	●	●	●	●	●											Test Circuits With Ohmmeter	S-3
Open Overload		●		●	●												Test Continuity of Overload	S-17A
Faulty Thermostat	●			●					●	●							Test continuity of Thermostat & Wiring	S-3
Faulty Transformer	●		●														Check control circuit w ith voltmeter	S-4
Shorted or Open Capacitor		●			●	●											Test Capacitor	S-15
Shorted or Grounded Compressor		●				●											Test Motor Windings	S-17B
Compressor Stuck	●					●											Use Test Cord	S-17D
Faulty Compressor Contactor	●	●			●	●											Test continuity of Coil & Contacts	S-7, S-8
2nd Stage Compressor Not Energized								●		●				●	●		Test Unloader Solenoid (2 stage compressor only)	S-17C
Faulty Ignition Control				●													Test Ignition Control	S-313
Open Control Circuit				●													Test Control Circuit w ith Voltmeter	S-4
Low Voltage		●				●	●										Test Voltage	S-1
Faulty Evap. Fan Motor				●									●				Repair or Replace	S-16
Shorted or Grounded Fan Motor					●											●	Test Motor Windings	S-16
Improper Cooling Anticipator							●		●								Check resistance of Anticipator	S-3
Shortage of Refrigerant							●	●					●	●			Test For Leaks, Add Refrigerant	S-103
Restricted Liquid Line							●	●					●	●			Replace Restricted Part	S-112
Dirty Air Filter								●		●	●		●				Inspect Filter-Clean or Replace	
Dirty Indoor Coil								●		●	●		●				Inspect Coil - Clean	
Not enough air across Indoor Coil								●		●	●		●				Speed Blow er, Check Duct Static Press	S-200
Too much air across Indoor Coil															●		Reduce Blow er Speed	S-200
Overcharge of Refrigerant						●	●					●			●	●	Recover Part of Charge	S-113
Dirty Outdoor Coil						●	●			●						●	Inspect Coil - Clean	
Noncondensibles							●			●						●	Recover Charge, Evacuate, Recharge	S-114
Recirculation of Condensing Air							●			●						●	Remove Obstruction to Air Flow	
Infiltration of Outdoor Air								●		●	●						Check Window s, Doors, Vent Fans, Etc.	
Improperly Located Thermostat						●			●								Relocate Thermostat	
Air Flow Unbalanced									●		●						Readjust Air Volume Dampers	
System Undersized								●		●							Refigure Cooling Load	
Broken Internal Parts												●					Replace Compressor	
Inefficient Compressor								●						●	●		Test Compressor Efficiency	S-104
High Pressure Control Open			●														Reset And Test Control	S-12
Unbalanced Pow er, 3PH		●				●	●										Test Voltage	
Wrong Type Expansion Valve						●	●	●		●			●	●			Replace Valve	
Expansion Device Restricted						●	●	●		●			●	●			Remove restriction or replace expansion device	
Expansion Valve Bulb Loose												●					Tighten Bulb Bracket	
Inoperative Expansion Valve						●		●					●				Check Valve Operation	S-110
Loose Hold-dow n Bolts												●					Tighten Bolts	

SERVICING

GAS HEATING ANALYSIS CHART

Complaint	No Heat			Unsatisfactory Heat						Test Method Remedy	See Service Procedure Reference
POSSIBLE CAUSE DOTS IN ANALYSIS GUIDE INDICATE "POSSIBLE CAUSE" SYMPTOM	System Will Not Start	Burner Won't Ignite	Burner Ignites-Locks Out	Burner Shuts Off prior to T'Stat being Satasfie	Short Cycles	Long Cycles	Soot and /or Fumes	To Much Heat	Not Enough Heat		
Pow er Failure	●									Test Voltage	S-1
Blow n Fuse	●									Test Voltage	S-4
Loose Connection	●									Check Wiring	S-2
Shorted or Broken Wires	●									Check Wiring	S-3
No Low Voltage	●									Check Transformer	S-4
Faulty Thermostat	●				●	●		●		Check Thermostat	S-3
Faulty Transformer	●									Check Transformer	S-4
Poor or High Resistance Ground			●							Measure Ground Resistance	S-313
Improper Heat Anticipator Setting					●	●		●	●	Adjust Heat Anticipator Setting	S-3
Improper Thermostat Location					●	●		●	●	Relocate Thermostat	
Faulty Limit or Roll Out Sw itch		●		●					●	Test Control	S-300,S-301,S-302
Faulty Flame Sensor			●							Test Flame Sensor	S-314
Faulty Ignition Control	●	●	●							Test Control	S-313
Gas Valve or Gas Supply Shut Off		●								Turn Valves to On Position	S-304
Faulty Induced Draft Blow er	●	●					●			Test Blow er	S-303
Dirty Flame Sensor, Low uA			●							Clean Flame Sensor	S-314
Flame Sensor not in Flame, Low uA			●							Test/Adjust Position of Flame Sensor	S-314
Faulty Gas Valve		●					●		●	Replace Gas Valve	S-304
No High Stage Heat (2 Stage Only)						●			●	Check Thermostat, Control, Gas Valve	S-3,S-313,S-304
Open Auxillary Limit		●		●						Reset Control	S-301
Improper Air Flow or Distribution				●					●	Check Duct Static	
Locking out on Main Limit			●	●					●	Check Controls, Gas Press., & Temp. Rise	S-300
Delayed Ignition							●			Test for Delayed Ignition	S-308
Flashback							●			Test for Flashback	S-309
Orifice Size							●	●	●	Check Orifices	S-306
Gas Pressure		●					●	●	●	Check Gas Pressure	S-307
Cracked Heat Exchanger							●			Check Burner Flames	S-302
Stuck Gas Valve		●					●	●		Replace Gas Valve	S-304
Furnace Undersized									●	Replace w ith Proper Size Furnace	
Faulty Pressure Sw itch		●	●	●						Test Pressure Sw itch	S-310
Blocked or Restricted Flue							●			Check Flue/Draw dow n Pressure	S-310
Open Roll Out Sw itch		●	●							Test Control	S-302
Bouncing On Pressure Sw itch			●						●	Test Negative Pressure	S-310

SERVICING

Table of Contents

S-1 Checking Voltage 35
S-2 Checking Wiring 36
S-3 Checking Thermostat, Wiring & Anticipator .. 36
S-3A Thermostat & Wiring 36
S-3B Cooling Anticipator 36
S-3C Heating Anticipator 36
S-4 Checking Transformer & Control Circuit 37
S-7 Checking Contactor and/or Relays 37
S-8 Checking Contactor Contacts 37
S-9 Checking Fan Relay Contact 38
S-12 Checking High Pressure Control 38
S-13 Checking Low Pressure Control 38
S-15 Checking Capacitor 38
S-15A Resistance Check 39
S-15B Capacitance Check 39
S-16C Checking ECM Motor Windings 40
S-16D Checking EEM Motors 40
S-17 Checking Compressor Windings 41
S-17A Resistance Test 41
S-17B Ground Test 42
S-17D Operation Test 42
S-18 Testing Crankcase Heater (optional item) 43
S-100 Refrigeration Repair Practice 43
S-101 Leak Testing 43
S-102 Evacuation 43
S-103 Charging 44
S-104 Checking Compressor Efficiency 45
S-105B Thermostatic Expansion Valve 45
S-106 Overfeeding 47
S-107 Underfeeding 47
S-108 Superheat 47
S-109 Checking Subcooling 49
S-110 Checking Expansion Valve Operation 49
S-111 Fixed Orifice Restriction Devices 49
S-112 Checking Restricted Liquid Line 50
S-113 Refrigerant Overcharge 50
S-114 Non-condensables 50
S-115 Compressor Burnout 50
S-200 Checking Extedrnal Static Pressure 51
S-201 Checking Temperature Rise 51
S-300 Testing Primary Limit 52
S-301 Testing Auxiliary Limit 52
S-302 Checking Flame Rollout Switch 53
S-303 Testing Inducer Motor 54
S-304 Testing Gas Valve 54
S-305 Checking Main Burners 55
S-306 Checking Orifices 55
S-307 Checking Gas Pressure 56
S-308 Checking For Delayed Ignition 57
S-309 Checking for Flashback 57
S-310 Checking Pressure Control 57
S-311 High Altitude Application 58
S-313 Testing Ignition Control Module 58
S-314 Checking Flame Sensor 60

⚠ WARNING

HIGH VOLTAGE!
Disconnect ALL power before servicing or installing this unit. Multiple power sources may be present. Failure to do so may cause property damage, personal injury or death.



SERVICING

S-1 CHECKING VOLTAGE

WARNING

HIGH VOLTAGE!
Disconnect ALL power before servicing or installing this unit. Multiple power sources may be present. Failure to do so may cause property damage, personal injury or death.



1. Remove doors, control panel cover, etc. from unit being tested.

With power ON:

WARNING

Line Voltage now present.

2. Using a voltmeter, measure the voltage across terminals L1 and L2 of the contactor for single phase units, and L3, for 3 phase units.
3. No reading - indicates open wiring, open fuse(s) no power or etc. from unit to fused disconnect service. Repair as needed.
4. With ample voltage at line voltage connectors, energize the unit.
5. Measure the voltage with the unit starting and operating, and determine the unit Locked Rotor Voltage.

   Locked Rotor Voltage is the actual voltage available at the compressor during starting, locked rotor, or a stalled condition. Measured voltage should be above minimum listed in chart below.

   To measure Locked Rotor Voltage attach a voltmeter to the run "R" and common "C" terminals of the compressor, or to the T_1 and T_2 terminals of the contactor. Start the unit and allow the compressor to run for several seconds, then shut down the unit. Immediately attempt to restart the unit while measuring the Locked Rotor Voltage.
6. Voltmeter should read within the voltage tabulation as shown. If the voltage falls below the minimum voltage, check the line wire size. Long runs of undersized wire can cause low voltage. If wire size is adequate, notify the local power company in regards to either low or high voltage.

Unit Supply Voltage		
Voltage	Min.	Max.
460	437	506
208/230	198	253

Three phase units require a balanced 3 phase power supply to operate. If the percentage of voltage imbalance exceeds 3% the unit must not be operated until the voltage condition is corrected.

$$\% \text{ Voltage Imbalance} = \frac{\text{Max. Voltage Deviation From Average Voltage}}{\text{Average Voltage}} \times 100$$

To find the percentage of imbalance, measure the incoming power supply.

L1 - L2 = 240V
L1 - L3 = 232V
L2 - L3 = 238V
Total 710V

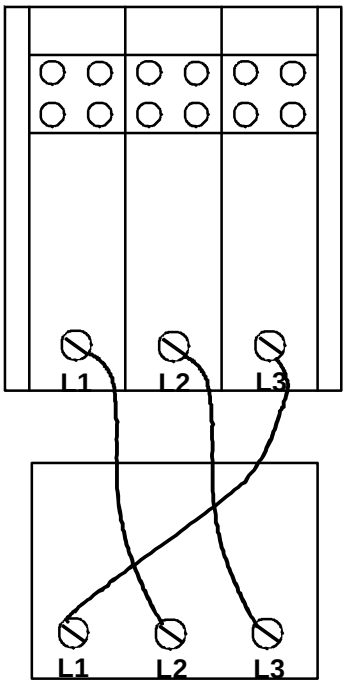
Avg. V = 710 / 3 = **236.7**

To find Max. deviation:
240 - 236.7 = +3.3
232 - 236.7 = **-4.7**
238 - 236.7 = +1.3

Max deviation was 4.7V

% Voltage Imbalance = 4.7 / 236.7 = **1.99%**

If the percentage of imbalance had exceeded 3%, it must be determined if the imbalance is in the incoming power supply or the equipment. To do this rotate the legs of the incoming power and retest voltage as shown below.



L1 - L2 = 240V
L1 - L3 = 227V
L2 - L3 = 238V

Rotate all 3 incoming legs as shown.

L1 - L2 = 227V
L1 - L3 = 238V
L2 - L3 = 240V

By the voltage readings we see that the imbalance rotated or traveled with the switching of the incoming legs. Therefore the imbalance lies within the incoming power supply.